

CHIBOUGAMAU CHAPAIS, QC, Canada

WMO: 718220

Lat: 49.7773N

Lon: 74.5303W

Elev: 1270

StdP: 14.03

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.5	-25.4	-39.2	0.6	-30.1	-34.0	0.8	-25.0	20.4	15.2	18.6	12.2	2.8	230	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	81.4	64.9	78.3	63.1	75.0	61.8	68.1	76.1	66.2	73.5	64.4	71.5	10.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	98.8	71.3	63.6	92.2	69.6	61.6	85.9	67.7	33.2	76.3	31.7	73.4	30.3	71.5	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.0	15.4	DB	-38.7	87.4	4.4	3.1	-41.9	89.6	-44.4	91.5	-46.9	93.2	-50.1	95.5	(4)
(5)			WB	-38.8	71.7	4.3	2.2	-41.9	73.3	-44.4	74.6	-46.8	75.8	-50.0	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.1	-0.5	2.3	14.6	29.8	46.2	57.3	61.8	59.3	51.4	38.9	24.5	9.3		
	DBStd	24.18	14.31	13.77	13.77	10.25	9.55	8.23	5.84	6.32	8.20	8.38	10.21	13.09		
	HDD50	7290	1565	1336	1097	606	191	25	0	5	79	359	765	1262		
	HDD65	11741	2030	1756	1561	1055	586	256	135	198	414	808	1215	1727		
	CDD50	1114	0	0	1	1	74	243	365	293	121	16	0	0		
	CDD65	90	0	0	0	0	4	24	35	21	6	0	0	0		
	CDH74	888	0	0	0	0	73	297	315	151	52	0	0	0		
	CDH80	161	0	0	0	0	14	68	55	17	7	0	0	0		
	WSAvg	7.5	7.0	7.3	7.8	8.1	7.8	7.4	7.1	6.9	7.4	8.0	7.8	7.0		
(15) Precipitation	PrecAvg	38.6	2.1	1.7	1.9	2.2	2.8	3.9	4.5	4.2	4.6	3.8	3.5	2.7		
	PrecMax	48.5	3.7	3.5	3.8	3.9	5.2	6.7	7.4	9.2	10.7	6.9	6.1	5.4		
	PrecMin	29.5	0.5	0.0	0.0	0.5	0.7	1.3	1.7	1.5	2.3	1.5	1.4	0.1		
	PrecStd	5.0	0.9	0.8	0.9	0.8	1.2	1.5	1.6	1.7	1.9	1.4	1.2	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	39.9	51.6	60.8	81.4	86.6	85.6	82.4	80.0	68.1	50.4	37.7		
		MCWB	38.6	37.0	42.8	46.1	62.3	66.2	67.4	66.8	67.2	57.7	46.0	36.4		
	2%	DB	32.6	33.4	43.4	55.3	74.9	81.1	80.8	78.2	74.1	61.6	45.6	33.8		
		MCWB	32.0	31.3	36.1	42.9	57.3	63.2	65.0	64.8	63.7	54.7	43.4	32.8		
	5%	DB	26.3	29.7	38.3	49.6	69.4	77.1	77.5	74.6	69.4	56.3	40.9	31.5		
		MCWB	24.9	27.7	33.4	40.2	54.6	61.4	63.0	63.2	60.9	51.9	39.1	30.6		
	10%	DB	21.5	24.5	34.4	44.7	63.9	72.8	73.9	71.4	65.3	51.6	37.2	28.1		
		MCWB	20.2	22.6	31.1	37.4	51.8	59.5	62.1	61.7	58.8	47.6	35.9	27.1		
	0.4%	WB	38.7	37.5	44.6	49.7	64.4	69.2	71.1	69.9	68.3	60.0	48.4	37.0		
		MCDB	39.0	39.2	47.2	56.1	76.6	79.9	79.7	77.6	75.5	65.5	49.6	37.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.0	31.6	37.4	44.2	59.6	66.6	68.1	67.3	65.5	55.8	43.4	32.9		
		MCDB	32.5	33.2	41.5	51.3	68.8	75.8	75.7	74.0	71.7	60.3	44.8	33.6		
	5%	WB	25.0	27.6	34.1	41.1	56.2	64.2	65.8	65.4	62.5	52.1	39.2	30.4		
		MCDB	26.1	29.4	37.7	48.2	66.2	72.6	73.1	72.0	68.2	56.1	40.5	31.2		
	10%	WB	20.3	22.8	31.3	38.2	53.3	61.7	63.8	63.3	59.3	48.4	35.9	27.0		
		MCDB	21.7	24.4	34.4	44.0	63.2	69.8	71.1	69.3	64.6	51.4	37.2	28.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	22.5	22.6	19.8	22.0	21.6	19.5	18.1	17.6	12.8	11.6	15.7		
		MCDBR	19.6	19.2	23.3	26.3	30.5	27.3	24.8	23.1	22.9	20.6	15.0	13.2		
		MCWBR	19.4	17.7	18.1	16.6	15.9	13.2	12.0	11.8	13.8	14.6	13.0	13.3		
		MCDBR	19.8	18.5	20.0	23.2	26.0	22.3	20.0	18.9	20.4	19.1	14.0	12.5		
	5% WB	MCWBR	19.8	17.6	16.4	16.2	15.4	12.5	11.4	11.3	13.6	14.7	13.1	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.266	0.279	0.327	0.337	0.357	0.372	0.355	0.347	0.305	0.285	0.244		
	taud		2.267	2.244	2.260	2.340	2.474	2.448	2.428	2.476	2.474	2.576	2.466	2.333		
	Ebn at Noon		255	280	296	291	292	286	279	280	270	267	244	244		
	Edh at Noon		22	29	35	36	33	34	34	31	28	22	19	18		
(44) All-Sky Solar Radiation	RadAvg		388	726	1203	1525	1659	1784	1621	1328	1007	566	283	233		
	RadStd		49	93	113	119	154	136	156	91	97	66	35	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+3.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page